piece of data: a time. That's what we mean by spacetime's being four-dimensional.

Kaluza proposed that in addition to left/right, back/forth, and up/down, *the universe actually has one more spatial dimension that, for some reason, no one has ever seen.* If correct, this would mean that there is another independent direction in which things can move, and therefore that we need to give four pieces of information to specify a precise location in space, and a total of five pieces of information if we also specify a time.

Okay; that's what the paper Einstein received in April 1919 proposed. The question is, Why didn't Einstein throw it away? We don't see another space dimension—we never find ourselves wandering aimlessly because a street, a cross street, and a floor number are somehow insufficient to specify an address—so why contemplate such a bizarre idea? Well, here's why. Kaluza realized that the equations of Einstein's general theory of relativity could fairly easily be extended mathematically to a universe that had one more space dimension. Kaluza undertook this extension and found, naturally enough, that the higher-dimensional version of general relativity not only included Einstein's original gravity equations but, because of the extra space dimension, also had extra equations. When Kaluza studied these extra equations, he discovered something extraordinary: the extra equations were none other than the equations Maxwell had discovered in the nineteenth century for describing the electromagnetic field! By imagining a universe with one new space dimension, Kaluza had proposed a solution to what Einstein viewed as one of the most important problems in all of physics. *Kaluza had found a framework that combined Einstein's original equations of general relativity with those of Maxwell's equations of electromagnetism.* That's why Einstein didn't throw Kaluza's paper away.

Intuitively, you can think of Kaluza's proposal like this. In general relativity, Einstein awakened space and time. As they flexed and stretched, Einstein realized that he'd found the geometrical embodiment of the gravitational force. Kaluza's paper suggested that the geometrical reach of space and time was greater still. Whereas Einstein realized that gravitational fields can be described as warps and ripples in the usual three space and one time dimensions, Kaluza realized that in a universe